

Compliance, Correction, and Convergence

Operational Specifications for the
Branch-Oriented Mathematical Architecture

[Author Name]

[Institution]

[email]

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Abstract

The Branch-Oriented Mathematical Architecture (BOMA), introduced in Paper I of this series, proposes an organizational paradigm for mathematical foundations in which every development is structured as a Branch Tree of locally consistent paths, every genuine choice is documented at an explicit Decision Point, and every new assumption is individually declared. Paper I established the framework’s theoretical architecture but left open a central operational question: what does it mean, precisely, for a mathematical development to be *compliant* with BOMA, and how does compliance manifest in practice?

This paper answers that question in four contributions. First, we formalize the notion of a *BOMA-compliant development* through six architectural properties — Traceability, Transparency, Auditability, Minimal-Brick Construction, Branchability, and Correctability — and establish their formal relationships to the structural components of Paper I.

Second, we enrich the binary classification of Paper I (Forced Step / Genuine Choice) into a four-way *Epistemic Status* taxonomy: Logical Necessity, Foundational Necessity, Architectural Necessity, and Methodological Choice. This taxonomy makes the Branchability property precise: only Methodological Choice Blocks generate Decision Points and potential Branches.

Third, we introduce two structural tools absent from Paper I. The *Conservativity Declaration* is a mandatory field in every Block’s structural record, making the distinction between zero-injection and non-zero-injection Blocks immediately legible. The *Internal Counter-Model* is an architectural diagnostic that characterizes the structural capacity of a single path from within, complementing the inter-path role of separation formulas in Paper I, and serves

as a structural prescription identifying precisely the injections needed for an Additive Correction.

Fourth, we distinguish two modes of correction: *Additive Correction*, appropriate when a development falls short of a claimed consequence, and *Branch Correction*, appropriate when a foundational choice requires revision. We demonstrate that the applicable mode is determined by the flaw type and the Epistemic Status of the affected Block.

All contributions are demonstrated through a worked example: the Pre-Numeric Kernel of the Transparent Mathematics project, an active constructive mathematics development that independently instantiates BOMA-compliance principles and provides the first empirical test of the framework’s structural predictions.

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1 Introduction

1.1 What Paper I Established and What Remains Open

The central contribution of BOMA Paper I [4] was architectural: it showed that the organizational paradigm of existing mathematical foundations — linear presentation of axioms and theorems, with foundational choices invisible, alternatives suppressed, and injection structure undeclared — is one paradigm among possible paradigms, and proposed a precise alternative. That alternative organized foundational development as a Branch Tree of locally consistent paths, required every genuine choice to be documented at an explicit Decision Point, and defined Architectural Transparency as the conjunctive property of four legibility conditions.

Paper I established this framework at the level of mathematical precision appropriate for a foundational proposal. What it did not establish was its operational content: the specific conditions a development must satisfy to be BOMA-compliant, the tools available for diagnosing structural gaps in an ongoing development, and the procedure for correcting discovered flaws without violating the immutability of completed Blocks.

These omissions were deliberate. Paper I was a framework paper; it defined the architecture without prescribing a protocol. The protocol is the subject of the present paper.

1.2 The Central Question

The question this paper addresses is simple to state and non-trivial to answer precisely:

Given a mathematical development in progress, what conditions must it satisfy to be BOMA-compliant, and how are violations corrected?

The simplicity is deceptive. Compliance is not a single condition but a conjunction of six, and the six conditions are not independent: the Minimal-Brick Construction property (P4) determines which Blocks generate Decision Points, which in turn determines the Branchability index (P5), which in turn determines where Branch Correction (P6) is applicable.

The correction question is equally non-trivial. BOMA-compliant Blocks are immutable once fixed; a development does not revise its past. When a flaw is discovered — as it inevitably will be in any serious mathematical project — the correction must be effected without in-place modification. This paper shows that two fundamentally different correction modes exist, that they apply in different circumstances, and that distinguishing them correctly is essential for maintaining the architectural integrity of the development.

1.3 The Role of the Practical Example

Alongside the formal development, this paper draws on a worked example: the Transparent Mathematics project, an active constructive mathematics initiative building mathematical foundations atomically from minimal traceable Bricks, with machine-verified formal content and explicit epistemic declarations [5].

The project was not designed as a BOMA implementation. It arose independently, from the same motivating concerns — opacity of foundational decisions, invisibility of alternatives, difficulty of tracing consequences — and converged on structural choices that instantiate BOMA-compliance principles in practice. The convergence is not coincidental: it reflects the fact that the structural requirements of transparent foundational construction, approached seriously from any direction, generate similar organizational discipline.

The project’s **ERRATA** document records a discovered flaw — an over-claim about the generative capacity of the Pre-Numeric Kernel — together with the counter-model that established it and the structural prescription for correction. This episode provides the first empirical test of the framework’s structural predictions, and its analysis motivates the distinction between Additive and Branch Correction developed in §6 and the formalization of the Internal Counter-Model in §7.

1.4 Contributions

This paper makes the following contributions.

Formalization of BOMA-Compliance. We define a BOMA-compliant development through six architectural properties (§4) and establish their formal relationships to the structural components of Paper I.

Four-Way Epistemic Status. We replace the binary Forced Step / Genuine Choice classification of Paper I with a four-way taxonomy — Logical Necessity, Foundational Necessity, Architectural Necessity, Methodological Choice — and show that the distinction carries structural consequences for the Branchability property (§2).

Conservativity as a Mandatory Field. We formalize Conservative and Non-Conservative Block extensions as a mandatory declaration in every Block’s structural record, and show its relationship to the injection weight of Paper I (§3).

Two Modes of Correction. We distinguish Additive Correction from Branch Correction, characterize the conditions under which each applies, and prove that the applicable mode is determined by the Epistemic Status of the Block in which the flaw originates (§6).

The Internal Counter-Model. We formalize the Counter-Model as an internal architectural diagnostic and show that it serves as a structural prescription identifying

precisely the injections needed for the next Additive Correction (§7).

The Architectural DAG and Convergent Blocks. We extend the Branch Tree of Paper I to the full Block-level Architectural DAG and introduce the Convergent Block — a Block appearing identically in two diverging paths — as the architectural criterion for foundational invariance (§5).

The Machine-Readable Block Record. We formalize the Block Summary Record as the infrastructure for mechanical Auditability (§8).

Worked Example. We analyze the Transparent Mathematics project’s Pre-Numeric Kernel as a complete instance of BOMA-compliance in practice (§9).

1.5 What This Paper Does Not Claim

This paper does not claim that the Transparent Mathematics project was designed as a BOMA implementation. It was not. The project is analyzed here as an independently arising instantiation of structural principles. The convergence between the project’s organizational discipline and BOMA’s structural requirements is evidence of the framework’s descriptive adequacy, not of the project’s theoretical dependence on it.

This paper does not claim that the six compliance properties are exhaustive or uniquely correct. They represent the properties that emerge from taking seriously the structural requirements of transparent foundational construction as identified in Paper I.

1.6 Paper Organization

Section 2 develops the four-way Epistemic Status taxonomy. Section 3 formalizes Conservativity as a mandatory declaration. Section 4 defines BOMA-compliance through six architectural properties. Section 5 introduces the Architectural DAG and the Convergent Block. Section 6 distinguishes Additive from Branch Correction. Section 7 formalizes the Internal Counter-Model. Section 8 establishes the Machine-Readable Block Record. Section 9 analyzes the Transparent Mathematics project. Section 10 relates the contributions to Paper I. Section 11 concludes.

2 Epistemic Status Taxonomy

2.1 Motivation

Paper I distinguished between two kinds of steps in an architectural development: *Forced Steps*, in which the content of the next Block is uniquely determined by prior cumulative content, and *Genuine Choices*, in which at least two distinct, logically

admissible, and architecturally non-equivalent continuations are available. This binary classification carries the essential structural load of Paper I.

The binary classification is adequate for Paper I's purposes, but it collapses distinctions that are epistemically significant. A step forced by the underlying logic is different in character from a step forced by the declared goals of the development: the former cannot be otherwise under any mathematical circumstances; the latter could be otherwise if the goal were revised. A step forced by the structural discipline of the framework is different again. The present section replaces the binary classification with a four-way *Epistemic Status* taxonomy that preserves the structural consequences of Paper I while making the source of necessity explicit in every case.

2.2 The Four Epistemic Statuses

Definition 2.1 (Epistemic Status). The *Epistemic Status* of a Block β is a mandatory declaration drawn from the following four-element set:

$$\text{ES}(\beta) \in \{\text{LN}, \text{FN}, \text{AN}, \text{MC}\}$$

where the four values are defined in Definitions 2.2–2.6 below.

Definition 2.2 (Logical Necessity — LN). A Block β has Epistemic Status *Logical Necessity* if its content is uniquely forced by the cumulative content of all ancestors under the declared object logic:

$$\text{ES}(\beta) = \text{LN} \iff \forall \varphi \in \text{Content}(\beta) : \widehat{\text{Content}(\overline{\text{Dep}(\beta)})} \vdash \varphi.$$

An LN Block has empty injection set ($\text{Inj}(\beta) = \emptyset$) and is therefore Conservative.

Definition 2.3 (Foundational Necessity — FN). A Block β has Epistemic Status *Foundational Necessity* if its content is not logically forced by prior content alone, but is forced by the conjunction of prior content and the declared goals of the development. An FN Block has non-empty injection set and is therefore Non-Conservative.

Remark 2.4. Foundational Necessity makes explicit a form of dependency often left implicit: the dependency of individual steps on the overall direction of the development. When a Block is declared FN, the declaration carries a commitment: the development's goal is declared, and the Block's relationship to that goal is stated. A reader who disputes the Block's content is directed to examine the goal declaration rather than the Block in isolation.

Definition 2.5 (Architectural Necessity — AN). A Block β has Epistemic Status *Architectural Necessity* if its content is required by the structural discipline of the

BOMA framework itself — by the architectural paradigm — rather than by the mathematical logic or the development’s mathematical goals. AN Blocks constitute the constitutional layer of a BOMA-compliant development.

Definition 2.6 (Methodological Choice — MC). A Block β has Epistemic Status *Methodological Choice* if its content represents a genuine choice among at least two distinct, logically admissible, and architecturally non-equivalent alternatives, none of which is forced by prior cumulative content, the development’s declared goals, or the architectural paradigm. An MC Block corresponds precisely to a Genuine Choice in the sense of Paper I, Definition 4.3.10, and generates a Decision Point.

Remark 2.7. The identification of a Block as MC may be *objective* (supported by a formal independence result) or *interpretive* (resting on the developer’s judgment), as established in Paper I, Remark 4.3.11. In both cases, the basis of identification must be declared in the Block’s Rationale component.

2.3 Structural Consequences

Proposition 2.8 (Injection Sets and Epistemic Status). *The following relationships hold:*

$$\begin{aligned} \text{ES}(\beta) = \text{LN} &\implies \text{Inj}(\beta) = \emptyset \\ \text{ES}(\beta) \in \{\text{FN}, \text{MC}\} &\implies \text{Inj}(\beta) \neq \emptyset \\ \text{ES}(\beta) = \text{AN} &\implies \text{Inj}(\beta) = \emptyset \text{ or } \text{Inj}(\beta) \neq \emptyset \end{aligned}$$

Proof. LN requires all content derivable from prior content, hence empty injection set. FN and MC require content not derivable from prior content alone, hence non-empty injection set. AN Blocks may or may not inject into the object-level theory depending on whether the architectural discipline is formalized within the same logical framework. $\square$

Proposition 2.9 (Decision Points and Epistemic Status). *A Block β is associated with a Decision Point if and only if $\text{ES}(\beta) = \text{MC}$.*

Proof. By Definition 2.6, an MC Block corresponds to a Genuine Choice, and by Paper I, Definition 4.3.10, every Genuine Choice gives rise to a Decision Point. Blocks with $\text{ES}(\beta) \in \{\text{LN}, \text{FN}, \text{AN}\}$ admit no genuine alternatives within their respective constraints. $\square$

Corollary 2.10 (Branchability Index). *A BOMA-compliant development $\mathcal{P}$ is branchable at Block β if and only if $\text{ES}(\beta) = \text{MC}$. The Branchability Index of $\mathcal{P}$ is:*

$$\text{BI}(\mathcal{P}) = |\{\beta \in \mathcal{P} \mid \text{ES}(\beta) = \text{MC}\}|.$$

2.4 The Mandatory Epistemic Status Declaration

Axiom 2.11 (Mandatory Epistemic Status Declaration). Every Block β in a BOMA-compliant development must include an explicit declaration of $\text{ES}(\beta)$, providing:

1. one of the four Epistemic Statuses of Definition 2.1;
2. a justification consistent with the chosen status;
3. the declaration at the time of Block introduction, not retroactively.

Proposition 2.12 (Compatibility with Paper I). *The four-way Epistemic Status taxonomy is a refinement of Paper I's binary classification:*

$$\begin{aligned} \text{Forced Step (Paper I)} &\equiv \text{ES}(\beta) \in \{\text{LN}, \text{FN}, \text{AN}\} \\ \text{Genuine Choice (Paper I)} &\equiv \text{ES}(\beta) = \text{MC}. \end{aligned}$$

All structural results of Paper I remain valid under the four-way taxonomy.

3 Conservativity as a Mandatory Field

3.1 Formal Definitions

Definition 3.1 (Conservative Extension). A Block β is a *conservative extension* of the development up to $\overline{\text{Dep}}(\beta)$ if every statement in its content is derivable from the cumulative content of its ancestors:

$$\text{IsConservative}(\beta) \iff \text{Inj}(\beta) = \emptyset.$$

A conservative Block adds no new foundational commitment.

Definition 3.2 (Non-Conservative Extension). A Block β is a *non-conservative extension* if $\text{Inj}(\beta) \neq \emptyset$. It introduces at least one statement not derivable from prior content and restricts the class of admissible models.

Definition 3.3 (Conservativity Field). The *Conservativity Field* of a Block β is the binary declaration:

$$\text{Cons}(\beta) \in \{\text{C}, \text{NC}\}$$

where C denotes Conservative and NC denotes Non-Conservative.

3.2 Mandatory Declaration and Relationships

Axiom 3.4 (Mandatory Conservativity Declaration). Every Block β in a BOMA-compliant development must include an explicit declaration of $\text{Cons}(\beta)$ satisfying:

1. **Accuracy:** the declared value matches the injection set;
2. **Contemporaneity:** the declaration is made at Block introduction;
3. **Verifiability:** supporting information for independent verification is provided.

Proposition 3.5 (Conservativity and Injection Weight). *For any Block β :*

$$\text{Cons}(\beta) = \text{C} \iff \omega(\beta) = 0 \quad \text{Cons}(\beta) = \text{NC} \iff \omega(\beta) > 0.$$

Proof. Immediate from Definitions 3.1–3.3 and Paper I, Definition 4.6.3: injection weight is zero iff the injection set is empty. $\square$

Proposition 3.6 (Conservativity and Epistemic Status). *The following relationships hold:*

$$\begin{aligned} \text{ES}(\beta) = \text{LN} &\implies \text{Cons}(\beta) = \text{C} \\ \text{ES}(\beta) \in \{\text{FN}, \text{MC}\} &\implies \text{Cons}(\beta) = \text{NC} \\ \text{ES}(\beta) = \text{AN} &\implies \text{Cons}(\beta) \in \{\text{C}, \text{NC}\}. \end{aligned}$$

Corollary 3.7 (Conservativity as a Necessary Condition for Branch Origins). *A Block β can be a Branch origin only if $\text{Cons}(\beta) = \text{NC}$.*

Proof. By Proposition 2.9, a Branch origin requires $\text{ES}(\beta) = \text{MC}$. By Proposition 3.6, $\text{ES}(\beta) = \text{MC}$ implies $\text{Cons}(\beta) = \text{NC}$. $\square$

Definition 3.8 (Conservativity Ratio). The *Conservativity Ratio* of a path $\mathcal{P}$ is:

$$\text{CR}(\mathcal{P}) = \frac{|\{\beta \in \mathcal{P} \mid \text{Cons}(\beta) = \text{C}\}|}{|\mathcal{P}|}.$$

Proposition 3.9 (Conservative Blocks Do Not Restrict Models). *Let β_k be a Conservative Block in a locally consistent path. Then every model satisfying the cumulative content of the preceding Blocks also satisfies the content of β_k .*

Proof. Since $\text{Cons}(\beta_k) = \text{C}$, every statement in $\text{Content}(\beta_k)$ is derivable from prior cumulative content. By soundness of the object logic, any model satisfying the premises satisfies the conclusion. $\square$

4 BOMA-Compliant Development

4.1 The Six Properties

Definition 4.1 (BOMA-Compliant Development). A mathematical development $\mathcal{P}$ is *BOMA-compliant* if it satisfies the following six architectural properties simultane-

ously.

P1 — Traceability. Every statement $\varphi \in \text{Th}(\mathcal{P})$ has a structurally determined provenance:

$$\text{Prov}(\varphi, \mathcal{P}) = \left(\{\beta_i \mid \varphi \text{ depends on } \text{Content}(\beta_i)\}, \right. \\ \left. \{\mathcal{D}_j \mid \varphi \text{ depends on } \text{Selection}(\mathcal{D}_j)\} \right)$$

recoverable from the structural record without external reconstruction. In practice, every formal entity carries a Global Unique Identifier of the form TYPE-BLOCK-SEQ.

P2 — Transparency. The development is architecturally transparent in the sense of Paper I, Definition 4.7.5, and additionally every Block declares $\text{ES}(\beta)$ (Axiom 2.11) and $\text{Cons}(\beta)$ (Axiom 3.4).

P3 — Auditability. Any researcher with access to the structural record can independently: (i) verify local consistency of any path; (ii) recover the provenance of any statement; (iii) identify the Decision Points and their documented alternatives; (iv) verify Conservativity declarations. *Mechanical Auditability* is achieved when the object logic supports automated proof checking; *Interpretive Auditability* is the minimum requirement.

P4 — Minimal-Brick Construction. Every Block satisfies the Minimal Injection Principle (Paper I, Axiom 4.6.6) with two additional disciplines:

- **Vocabulary-Axiom Decoupling:** the introduction of a vocabulary item and the introduction of axioms governing it occupy separate Blocks.
- **Single-Purpose Commitment:** each Non-Conservative Block introduces exactly one logically independent core concept.

P5 — Branchability. The Branchability Index $\text{BI}(\mathcal{P})$ is accurately computed and every MC Block is associated with a fully documented Decision Point satisfying Paper I, Axiom 4.3.4. Non-MC Blocks are not Branch origins, and the framework imposes no requirement that every Block be branchable.

P6 — Correctability. Every discovered flaw is remedied through one of two architecturally admissible correction procedures (Additive or Branch Correction, as defined in §6), and no completed Block is modified in place. Completed Blocks are immutable; their `ErratumStatus` field is the only field that may be updated after the Block is marked Immutable.

4.2 Formal Definition

Definition 4.2 (BOMA-Compliance — Formal). A mathematical development $\mathcal{P}$ is *BOMA-compliant* if and only if:

1. $\mathcal{P}$ is architecturally transparent (Paper I, Definition 4.7.5);
2. every Block declares $\text{ES}(\beta)$ satisfying Axiom 2.11;
3. every Block declares $\text{Cons}(\beta)$ satisfying Axiom 3.4;
4. every Block satisfies P4 (Minimal-Brick Construction);
5. $\text{BI}(\mathcal{P})$ is accurately computed and every MC Block has a fully documented Decision Point;
6. every flaw is remedied through an admissible correction procedure with no in-place modification.

4.3 Compliance Hierarchy and Relations to Paper I

Proposition 4.3 (BOMA-Compliance Strengthens Architectural Transparency). *Every BOMA-compliant development is architecturally transparent (Paper I, Definition 4.7.5). The converse does not hold: Architectural Transparency does not require Epistemic Status, Conservativity, Vocabulary-Axiom Decoupling, or the ERRATA mechanism.*

Proposition 4.4 (BOMA-Compliance and Local Consistency). *Every locally consistent path in a BOMA-compliant development remains locally consistent after any Additive Correction, and after any Branch Correction that selects a locally consistent alternative.*

Proof. Additive Correction introduces new Blocks that are locally consistent in isolation. No completed Block is modified in place; the original path’s local consistency is unaffected. Branch Correction opens a new path whose local consistency is assessed independently by Paper I, Principle 4.5.5. $\square$

5 The Architectural DAG and Convergent Blocks

5.1 The Architectural DAG

Definition 5.1 (Architectural DAG). The *Architectural DAG* of a BOMA-compliant development is a directed acyclic graph $\mathcal{G} = (V, E)$ where:

- V is the set of all Blocks appearing in any locally consistent path of the development;
- E contains a directed edge (β, β') iff $\beta \in \text{Dep}(\beta')$ for some path.

Acyclicity follows from Paper I, Axiom 4.2.4, together with the global identifier ordering (Block $\beta \in \text{Dep}(\beta')$ implies $\text{Id}(\beta) < \text{Id}(\beta')$).

Definition 5.2 (Branch Tree as Projection). The Branch Tree $\mathcal{T}$ of Paper I, Definition 4.4.12, is the *Decision-Level Projection* of $\mathcal{G}$: the sub-DAG induced by Decision Blocks alone, with transitive edges between consecutive Decision Blocks.

Remark 5.3. The projection from $\mathcal{G}$ to $\mathcal{T}$ is lossy: it retains the decision-level structure but discards the Block-level structure between Decision Points. Two developments with identical Branch Trees may have very different Architectural DAGs.

5.2 Convergent Blocks

Definition 5.4 (Convergent Block). A Block β is a *Convergent Block* with respect to locally consistent paths $\mathcal{P}_1$ and $\mathcal{P}_2$ diverging at Decision Point $\mathcal{D}$ if:

1. $\beta \in \mathcal{P}_1$ and $\beta \in \mathcal{P}_2$;
2. $\text{Content}(\beta)$ is identical in both paths;
3. $\beta \notin \mathcal{P}_1 \wedge \mathcal{P}_2$ (the Block is not part of the common ancestry).

Condition 3 distinguishes Convergent Blocks from ordinary shared ancestors: a Convergent Block is reached independently after the divergence at $\mathcal{D}$.

Definition 5.5 (Convergence Point and Depth). A *Convergence Point* in $\mathcal{G}$ is a vertex $\beta \in \text{Conv}(\mathcal{G})$ that is a Convergent Block with respect to at least two paths. The *convergence depth* of β with respect to $\mathcal{P}_1$ and $\mathcal{P}_2$ is:

$$\text{CD}(\beta, \mathcal{P}_1, \mathcal{P}_2) = \min(|\mathcal{P}_1 \setminus (\mathcal{P}_1 \wedge \mathcal{P}_2)|, |\mathcal{P}_2 \setminus (\mathcal{P}_1 \wedge \mathcal{P}_2)|).$$

5.3 Path-Independence and Mathematical Significance

Theorem 5.6 (Path-Independence of Convergent Blocks). *Let β be a Convergent Block with respect to locally consistent paths $\mathcal{P}_1$ and $\mathcal{P}_2$ diverging at $\mathcal{D}$. Then:*

1. *The content of β is architecturally present and structurally valid in both paths.*
2. *The content of β is path-independent with respect to $\mathcal{D}$: the selection at $\mathcal{D}$ was not sufficient to prevent β 's content from being reached in either path.*

3. $\text{Content}(\beta) \subseteq \text{Th}(\mathcal{P}_1) \cap \text{Th}(\mathcal{P}_2)$.

4. $\mathcal{D}$ is non-trivial, and β 's content is precisely what $\mathcal{D}$'s non-triviality does not affect.

Proof. (1) By Definition 5.4, β appears in both paths with identical content; whether Conservative or Non-Conservative, it is architecturally valid in both.

(2) Since $\beta \notin \mathcal{P}_1 \wedge \mathcal{P}_2$, it was reached after the divergence. Both paths arrived at β with identical content despite different post-Decision histories. The content is therefore not determined by the selection at $\mathcal{D}$.

(3) $\text{Content}(\beta) \subseteq \text{Th}(\mathcal{P}_1)$ and $\text{Content}(\beta) \subseteq \text{Th}(\mathcal{P}_2)$ follow from (1); the intersection claim follows.

(4) Paper I, Definition 4.5.7 establishes non-triviality from the divergence of locally consistent paths. The content of β , being common to both, falls outside the scope of $\mathcal{D}$'s non-triviality. $\square$

Definition 5.7 (Decision-Point Invariant). The *Decision-Point Invariant* of $\mathcal{D}$ is:

$$\text{Inv}(\mathcal{D}) = \left\{ \beta \mid \beta \text{ is a Convergent Block w.r.t. some } \mathcal{P}_1, \mathcal{P}_2 \text{ diverging at } \mathcal{D} \right\}.$$

Proposition 5.8 (Complementarity of Invariants and Separation Formulas). *For any non-trivial $\mathcal{D}$ with paths $\mathcal{P}_1, \mathcal{P}_2$:*

$$\text{Content}(\beta) \in \text{Inv}(\mathcal{D}) \implies \text{Content}(\beta) \notin \text{Sep}(\mathcal{D}).$$

The content of Convergent Blocks and the separation formulas of Paper I are disjoint.

Proof. A separation formula $\varphi \in \text{Sep}(\mathcal{D})$ satisfies $\mathcal{P}_1 \vdash \varphi$ and $\mathcal{P}_2 \vdash \neg\varphi$. If $\varphi \in \text{Content}(\beta)$ for a Convergent Block β , then $\mathcal{P}_2 \vdash \varphi$, contradicting $\mathcal{P}_2 \vdash \neg\varphi$ given local consistency of $\mathcal{P}_2$. $\square$

Corollary 5.9 (Architectural Portrait of a Decision Point). *The complete architectural portrait of a non-trivial $\mathcal{D}$ with respect to $\mathcal{P}_1$ and $\mathcal{P}_2$ consists of: the common ancestry $\mathcal{P}_1 \wedge \mathcal{P}_2$; the separation formulas $\text{Sep}(\mathcal{D})$ (what $\mathcal{D}$ makes different); the Decision-Point Invariant $\text{Inv}(\mathcal{D})$ (what $\mathcal{D}$ leaves unchanged); and the injection divergence $\Delta_{\text{Inj}}(\mathcal{P}_1, \mathcal{P}_2)$ (Paper I, Definition 6.3.5).*

6 Two Types of Correction

6.1 The ERRATA Mechanism

Definition 6.1 (ERRATA Entry). An *ERRATA Entry* for a discovered flaw is a formal record:

$$\text{ERR} = (\text{Id}, \text{Location}, \text{Nature}, \text{Witness}, \text{Correction})$$

where:

- $\text{Nature} \in \{\text{LogicalInconsistency}, \text{OverClaim}, \text{MisclassifiedEpistemic}, \text{ViolatedMinimalInjection}\}$
- Witness is a formal demonstration of the flaw: a derivation of $\perp$, a counter-model, or a derivation of a supposedly injected statement
- $\text{Correction} \in \{\text{Additive}, \text{Branch}\}$

Axiom 6.2 (Mandatory ERRATA Logging). Every discovered flaw must be formally logged as an ERRATA Entry before any correction is introduced. The flawed Block receives an `ErratumStatus` field pointing to $\text{Id}(\text{ERR})$. The ERRATA Entry is itself immutable once logged.

6.2 The Two Correction Modes

Definition 6.3 (Additive Correction). An *Additive Correction* introduces new Blocks into the current path without opening a Branch and without modifying any existing Block. If the current path is $\mathcal{P} = (\beta_0, \dots, \beta_n)$, an Additive Correction produces:

$$\mathcal{P}^+ = (\beta_0, \dots, \beta_n, \gamma_1, \gamma_2, \dots, \gamma_k)$$

where $\gamma_1, \dots, \gamma_k$ are new Blocks addressing the flaw.

Definition 6.4 (Branch Correction). A *Branch Correction* opens a new Branch at the Decision Point associated with the flawed MC Block, selecting a different option. If the flawed Block β_j corresponds to Decision Point $\mathcal{D}_j$ with selection O_s , a Branch Correction selects some $O_i \neq O_s$ and opens:

$$\mathcal{B}_i = (\beta_0, \dots, \beta_{j-1}, \gamma_j^i, \gamma_{j+1}^i, \dots)$$

where $(\beta_0, \dots, \beta_{j-1})$ is the branching prefix inherited exactly.

6.3 Correction Mode Determination

Theorem 6.5 (Correction Mode Determination). *The applicable correction mode is determined by the flaw type and the Epistemic Status of the affected Block as follows:*

<i>Flaw Type</i>	$ES(\beta)$	<i>Mode</i>
OverClaim	<i>Any</i>	<i>Additive</i>
LogicalInconsistency	LN or FN	<i>Additive</i>
LogicalInconsistency	MC	<i>Branch</i>
MisclassifiedEpistemic: LN/FN/AN declared, MC correct	<i>Any</i>	<i>Branch</i>
MisclassifiedEpistemic: MC declared, LN/FN/AN correct	<i>Any</i>	<i>Additive</i>
ViolatedMinimalInjection	<i>Any</i>	<i>Additive</i>

Proof sketch. Over-Claim: The flaw is in attributed consequences, not in any Block's content. Existing Blocks are sound; the path must be extended with the missing injections. No Branch is needed.

Logical Inconsistency, LN/FN: Content was forced by logic or goals. Inconsistency traces to earlier injections that combine to produce a contradiction. Corrective Blocks address the source additively.

Logical Inconsistency, MC: The chosen option at a Decision Point led to inconsistency. A different option must be selected: Branch Correction opens the alternative path.

Misclassified LN/FN/AN when MC is correct: A genuine alternative was suppressed. The correction opens the Decision Point that was previously invisible: Branch Correction applies.

Misclassified MC when LN/FN/AN is correct: A spurious Decision Point was declared. A supplementary Additive Block formally establishes that the content was forced.

Violated Minimal Injection: A Block injected more than necessary. A corrective Additive Block replaces the strong injection with the weaker sufficient statement. $\square$

Proposition 6.6 (Incoherence of Wrong-Mode Correction). *Applying Branch Correction to a flaw that requires Additive Correction, or vice versa, produces architecturally incoherent results.*

Proof. Branch Correction applied to an Over-Claim changes the foundational option selected but does not supply the missing injections; the over-claim persists. Additive

Correction applied to a Logical Inconsistency in an MC Block extends an already-invalid path (by explosion) without resolving the inconsistency at its source. In each case the flaw is not addressed by the applied mode. $\square$

7 The Internal Counter-Model

7.1 Formal Definitions

Definition 7.1 (Internal Counter-Model). Let $\mathcal{P}$ be a locally consistent BOMA-compliant path and φ a formula in the path's declared object logic. An *internal counter-model* for φ relative to $\mathcal{P}$ is a model $\mathcal{M}$ satisfying:

1. *Compatibility*: $\mathcal{M} \models \widehat{\text{Content}}(\mathcal{P})$;
2. *Falsification*: $\mathcal{M} \not\models \varphi$.

The existence of $\mathcal{M}$ establishes $\mathcal{P} \not\vdash \varphi$.

7.2 Existence and Non-Derivability

Theorem 7.2 (Counter-Model Existence). *Let $\mathcal{P}$ be locally consistent. For any formula φ :*

$$\mathcal{P} \not\vdash \varphi \iff \exists \mathcal{M} : \mathcal{M} \models \widehat{\text{Content}}(\mathcal{P}) \wedge \mathcal{M} \not\models \varphi.$$

Proof. Right-to-left: soundness of the object logic. Left-to-right: for logics with a completeness theorem (classical first-order logic via Gödel's completeness theorem [1]; intuitionistic logic via Kripke completeness [2]), if $\mathcal{P} \not\vdash \varphi$ then $\widehat{\text{Content}}(\mathcal{P}) \cup \{\neg\varphi\}$ is consistent and has a model $\mathcal{M}$. $\square$

7.3 Counter-Models as Architectural Diagnostics

Theorem 7.3 (Counter-Model as Structural Prescription). *Let $\mathcal{M}$ be an internal counter-model for φ relative to locally consistent $\mathcal{P}$. Then:*

1. **Consistency confirmation**: $\mathcal{P}$ is locally consistent.
2. **Structural deficit identification**: The conditions satisfied by $\mathcal{M}$ but incompatible with φ constitute the precise structural deficit of $\mathcal{P}$ with respect to φ .
3. **Additive Correction prescription**: The minimal set of injections that would rule out $\mathcal{M}$ constitutes the minimal Additive Correction required to make φ derivable.

4. **Non-branching character:** *The counter-model does not compare $\mathcal{P}$ to any other path; it characterizes $\mathcal{P}$ from within.*

Proof. (1) $\mathcal{M} \models \widehat{\text{Content}}(\mathcal{P})$ establishes satisfiability, hence consistency. (2) The structural deficit $S(\mathcal{M}, \varphi)$ consists of conditions in $\mathcal{M}$ not ruled out by $\widehat{\text{Content}}(\mathcal{P})$ that permit φ to be falsified. (3) The negations of elements of $S(\mathcal{M}, \varphi) \setminus \widehat{\text{Content}}(\mathcal{P})$ are the required injections; minimality follows from the maximally informative counter-model construction. (4) The construction requires only $\mathcal{P}$ and φ ; no other path is referenced. $\square$

Proposition 7.4 (Counter-Model Confirms Local Consistency). *If an internal counter-model for any φ relative to $\mathcal{P}$ exists, then $\mathcal{P}$ is locally consistent.*

Proof. $\mathcal{M} \models \widehat{\text{Content}}(\mathcal{P})$ implies satisfiability of $\widehat{\text{Content}}(\mathcal{P})$; satisfiability implies consistency. $\square$

7.4 Counter-Models vs Separation Formulas

Proposition 7.5 (Complementarity). *The internal counter-model and the separation formula of Paper I are complementary tools:*

Property	Counter-Model	Separation Formula
Scope	Single path $\mathcal{P}$	Two paths $\mathcal{P}_1, \mathcal{P}_2$
Establishes	$\mathcal{P} \not\models \varphi$	$\mathcal{P}_1 \vdash \varphi, \mathcal{P}_2 \vdash \neg\varphi$
Primary use	Diagnostic + Prescription	Comparative analysis

Neither subsumes the other; both are necessary for a complete account of derivability in the Foundational Space.

Definition 7.6 (Canonical Counter-Model). A counter-model $\mathcal{M}$ for φ relative to $\mathcal{P}$ is *canonical* if it is both minimal (no proper sub-model is also a counter-model) and maximally informative (the structural deficit $S(\mathcal{M}, \varphi)$ is minimal among all counter-models for φ relative to $\mathcal{P}$).

Corollary 7.7 (Pre-Verification of Additive Correction). *An Additive Correction prescribed by a canonical internal counter-model is automatically Minimal-Injection-compliant: every prescribed injection is non-derivable from prior content, and the set of injections is minimal.*

8 The Machine-Readable Block Record

8.1 The Block Record Schema

Definition 8.1 (Block Summary Record). The *Block Summary Record* of Block β is a structured document with the following fields:

Listing 1: Block Summary Record Schema

```
# Group I: Identity
ID:          TYPE-BLOCK-SEQ
Title:       [Short descriptive name]
Status:      Draft | Immutable

# Group II: Architectural Character
EpistemicStatus:      LN | FN | AN | MC
EpistemicJustification: [text or reference]
Conservativity:       Conservative | NonConservative
ConservativityBasis:  [text or reference]

# Group III: Dependency and Content
Depends_on:  [TYPE-BLOCK-SEQ, ...]
Introduces:  [VOC-BLOCK-SEQ, ...]
Exports:     [TYPE-BLOCK-SEQ, ...]
Injects:     [AXM-BLOCK-SEQ, ...]
Derives:     [DEF/THM-BLOCK-SEQ, ...]

# Group IV: Decision and Correction
DecisionPoint:
  ID:          DP-BLOCK-SEQ | null
  Options:     [...] | null
  Selection:   [...] | null
  SelectionRationale: [text] | null
  IndependenceBasis: Objective | Interpretive | null
  UnChosenOptions: [...] | null
ErratumStatus:
  ID:          ERR-SEQ | null
  Nature:      [...] | null
  CorrectionMode: Additive | Branch | null
  CorrectionRef: [...] | null

# Group V: Formal Payload
FormalPayload:
```

Language:	Lean4 Coq Agda Isabelle Informal
Source:	file:path inline null
VerificationStatus:	Compiled ProofChecked Reviewed Unverified

Axiom 8.2 (Mandatory Block Record). Every Block β in a BOMA-compliant development must have a Block Summary Record satisfying:

1. **Completeness:** all fields populated;
2. **Accuracy:** every field accurately reflects the Block’s structural situation at introduction;
3. **Contemporaneity:** produced at Block introduction, not retroactively;
4. **Consistency:** structural declarations consistent with the formal payload.

8.2 Block Records and BOMA-Compliance

Proposition 8.3 (Block Records Establish Compliance). *A BOMA-compliant development in which every Block has a complete and accurate Block Summary Record automatically satisfies all six compliance properties of Definition 4.1.*

Proof. **P1:** Depends_on and Injects enable provenance tracing. **P2:** Epistemic-Status, Conservativity, and all legibility fields implement Transparency. **P3:** The record supports all four auditing operations; VerificationStatus records mechanical verification. **P4:** The separation of Injects and Derives enforces Minimal Injection; Vocabulary-Axiom Decoupling requires Introduces and Injects to appear in separate Blocks. **P5:** DecisionPoint fields provide Branchability documentation. **P6:** Status enforces immutability; ErratumStatus implements the ERRATA mechanism. □

8.3 Automated Compliance Checking

Definition 8.4 (Compliance Checker). A *Compliance Checker* verifies the following conditions from the Block Records:

- C1** Every Block has a complete Record.
- C2** The Depends_on graph is acyclic.
- C3** Conservativity is consistent with Injects.
- C4** EpistemicStatus is consistent with Conservativity (Proposition 3.6).
- C5** Every MC Block has a fully populated DecisionPoint group.

- C6** Every non-null `ErratumStatus.ID` refers to an existing `ERRATA` entry.
- C7** `Exports` is the union of `Introduces`, `Injects`, `Derives`; all IDs are unique; every `Depends_on` ID is in `Exports` of a prior Block.
- C8** No `Immutable` Block has any field other than `ErratumStatus` modified after the status was assigned.

All checks C1–C8 are decidable from a finite set of Block Records.

9 Worked Example: The Pre-Numeric Kernel

9.1 Overview

The Pre-Numeric Kernel of the Transparent Mathematics project spans sixteen Bricks organized into two groups: the Zero-Group Constitution (Bricks 000–006), containing meta-mathematical rules governing the development’s conduct; and the Pre-Numeric Kernel Proper (Bricks 007–015), containing formal mathematical content formalized in Lean 4 [6, 3].

The project’s organizational discipline — atomic Bricks, vocabulary-axiom decoupling, explicit epistemic declarations, immutability, formal `ERRATA` logging — instantiates the six compliance properties independently of `BOMA`. The convergence between the project’s structure and `BOMA`’s requirements is evidence of the framework’s descriptive adequacy.

9.2 Compliance Assessment

P1 (Traceability). Substantially satisfied. The Global Unique Identifier system (`TYPE-BLOCK-SEQ`) enables provenance tracing for every exported entity. The dependency graph of the Lean 4 identifiers is tracked by the proof assistant itself.

P2 (Transparency). Largely satisfied. The `Necessity Type` declarations in the project’s machine-readable summaries map exactly onto the four-way Epistemic Status taxonomy, and `Conservativity` declarations are present in all Bricks 007–015. Minor gap: Bricks 000–006 use an older format lacking the full Block Record schema of Definition 8.1.

P3 (Auditability). Mechanical Auditability achieved for Bricks 007–015 (Lean 4 compilation and proof checking). Interpretive Auditability for Bricks 000–006. The Compliance Checker of Definition 8.4 has not yet been implemented.

P4 (Minimal-Brick Construction). Fully satisfied. Vocabulary-Axiom Decoupling is rigorously maintained: every vocabulary item (Bricks 008, 010, 011, 013) is introduced in a dedicated Brick before any axiom governs it. Single-Purpose Commitment is observed throughout.

P5 (Branchability). Satisfied. With both Brick 007 (logic choice) and Brick 011 (primitive positive distinction) classified as MC:

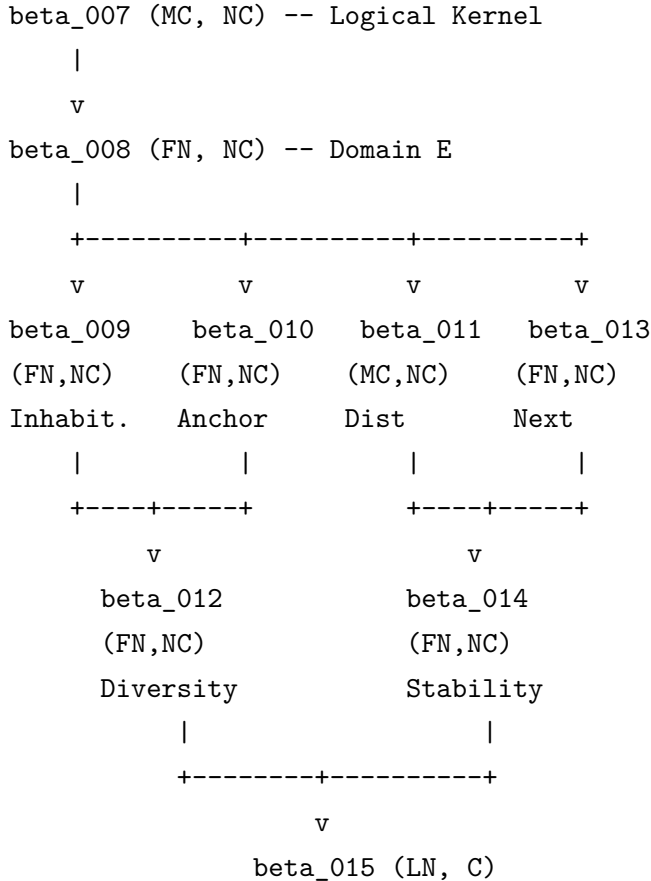
$$\text{BI}(\mathcal{P}_{\text{PNK}}) = 2.$$

Both Decision Points are documented with options, selections, rationales, and independence bases.

P6 (Correctability). Satisfied through ERR-001's Additive Correction. No in-place modification was made; the flaw is formally logged with a canonical counter-model witness; the correction prescription is specified.

9.3 The Architectural DAG

The full Architectural DAG of the Pre-Numeric Kernel (Bricks 007–015) has the following structure:



Derived Predicate

Structural properties: 9 vertices, 13 edges, maximum in-degree 4 (β_{014}), maximum out-degree 4 (β_{008}), 1 Conservative node (β_{015}), 1 Convergence Point (candidate: $D(x)$ predicate in future Branches).

Remark 9.1. The DAG is non-linear at Brick 008 (out-degree 4) and Brick 014 (in-degree 4) as a direct consequence of Vocabulary-Axiom Decoupling: the domain $\mathbf{E}$ is introduced once and used by four subsequent Bricks (Proposition 3.6).

9.4 ERR-001: Full BOMA Analysis

Flaw Classification. OverClaim: the Milestone-001 documentation asserted infinite generative capacity, but this is not in $\text{Th}(\mathcal{P}_{\text{PNK}})$.

Correction Mode (Theorem 6.5). OverClaim always requires Additive Correction regardless of Epistemic Status. ✓

Counter-Model Analysis. The counter-model $\mathcal{M}_{\text{ERR}}$:

$$\mathbf{E} = \{\bullet\}, \quad o = \bullet, \quad \text{Dist}(x, y) = \top, \quad \text{Next}(x) = x.$$

Compatibility: all axioms AXM-009-001, AXM-012-001, AXM-014-001 are satisfied. ✓

Falsification: no infinite sequence of pairwise distinct elements exists in a one-element universe. ✓

Consistency confirmation (Proposition 7.4): $\mathcal{P}_{\text{PNK}}$ is locally consistent. ✓

Structural Deficit and Prescription. Two conditions permit $\mathcal{M}_{\text{ERR}}$:

$$\begin{aligned} \text{Dist}(\bullet, \bullet) &= \top \quad (\text{reflexivity of Dist not ruled out}), \\ \text{Next}(\bullet) &= \bullet \quad (\text{fixed-point of Next not ruled out}). \end{aligned}$$

Minimal Additive Correction (Theorem 7.3):

$$\begin{aligned} \psi_{\text{irr}} &: \forall x : \mathbf{E}. \neg \text{Dist}(x, x), \\ \psi_{\text{inj}} &: \forall x, y : \mathbf{E}. \text{Next}(x) = \text{Next}(y) \rightarrow x = y. \end{aligned}$$

Canonical Character. $\mathcal{M}_{\text{ERR}}$ is minimal (one-element domain, no proper sub-models) and maximally informative (identifies exactly two structural deficits). By Definition 7.6, it is a canonical counter-model. ✓

Pre-Verification (Corollary 7.7). The canonical counter-model certifies Minimal-
Injection compliance of the Additive Correction before any new Brick is introduced.
✓

9.5 Block Records: Three Representative Examples

Example 9.2 (Brick_007 — Methodological Choice, Non-Conservative).

ID: AXM-007-LOG Title: Intuitionistic Logical Kernel Status: Immutable EpistemicStatus: MethodologicalChoice EpistemicJustification: > Classical first-order logic was a genuine alternative. Intuitionistic logic was selected to require explicit construction for all existential claims. Independence basis: objective (Goedel-Gentzen negative translation). Conservativity: NonConservative ConservativityBasis: > Introduces AXM-007-001 and AXM-007-002, neither derivable from prior content (there is no prior content). Depends_on: [] Introduces: [] Exports: [AXM-007-001, AXM-007-002] Injects: [AXM-007-001, AXM-007-002] Derives: [] DecisionPoint: ID: DP-007-001 Options: [Classical FOL, Intuitionistic MSFOL, Relevant] Selection: Intuitionistic MSFOL SelectionRationale: > Preserves constructive existence; prevents non-constructive existence proofs from infiltrating the kernel. IndependenceBasis: Objective UnChosenOptions: [Classical FOL, Relevant logic] ErratumStatus: ID: null FormalPayload:
--

Language:	Lean4
Source:	Formalization/PreNumericKernel.lean
VerificationStatus:	Compiled

Example 9.3 (Brick_012 — Foundational Necessity, Non-Conservative).

```

ID:      AXM-012-001
Title:   Anchor-Relative Diversity Axiom
Status:  Immutable

EpistemicStatus:      FoundationalNecessity
EpistemicJustification: >
    Given the development's goal of structural diversity, at least
    two distinguishable elements are needed. Anchor-Relative
    formulation is minimal; no weaker alternative achieves the
    structural goal.
Conservativity:       NonConservative
ConservativityBasis: >
    AXM-012-001 not derivable from prior vocabulary and inhabitation
    axioms. Consistency witness: two-element model {o, a} with
    Dist(a, o) = True.

Depends_on: [VOC-008-001, AXM-009-001, VOC-010-001, VOC-011-001]
Introduces: []
Exports: [AXM-012-001]
Injects: [AXM-012-001]
Derives: []

DecisionPoint:
    ID: null

ErratumStatus:
    ID: null

FormalPayload:
    Language: Lean4
    Source: Formalization/PreNumericKernel.lean
    VerificationStatus: Compiled

```

Example 9.4 (Brick_015 — Logical Necessity, Conservative).

```

ID:      DEF-015-001
Title:   Derived Predicate and Generative Consequences
Status:  Immutable

```

EpistemicStatus:	LogicalNecessity
EpistemicJustification:	> DEF-015-001 is a pure macro (eliminable). THM-015-001, THM-015-002, THM-015-003 are strict logical consequences of AXM-012-001 and AXM-014-001, verified by Lean4.
Conservativity:	Conservative
ConservativityBasis:	> No new axioms or vocabulary. All content derivable from prior Bricks. Lean4 proofs require no additional axioms.
Depends_on:	[VOC-011-001, AXM-012-001, VOC-013-001, AXM-014-001]
Introduces:	[]
Exports:	[DEF-015-001, THM-015-001, THM-015-002, THM-015-003]
Injects:	[]
Derives:	[DEF-015-001, THM-015-001, THM-015-002, THM-015-003]
DecisionPoint:	ID: null
ErratumStatus:	ID: ERR-001 Nature: OverClaim CorrectionMode: Additive CorrectionRef: [AXM-MILESTONE002-IRREF, AXM-MILESTONE002-INJ]
FormalPayload:	Language: Lean4 Source: Formalization/PreNumericKernel.lean VerificationStatus: ProofChecked

9.6 Architectural Insights from the Example

The Pre-Numeric Kernel analysis generates five architectural observations that extend the theoretical development of preceding sections.

Observation 9.5 (Vocabulary Bricks Are Systematically Non-Conservative). Every Brick introducing a vocabulary item (Bricks 008, 010, 011, 013) is Non-Conservative: vocabulary introduction expands the signature, enabling new formulas even before any axiom governs the new symbol. This is a general pattern: *vocabulary introduction is always Non-Conservative*.

Observation 9.6 (The First Conservative Threshold). Brick 015 is the first Conser-

vative Brick in the object-level development, appearing after 8 Non-Conservative object-level Bricks. This *First Conservative Threshold* marks the moment at which the development has accumulated sufficient foundational content to generate non-trivial derived consequences.

Observation 9.7 (Counter-Models Are More Informative Than Non-Derivability Proofs). An abstract proof that φ_∞ is not derivable from $\mathcal{P}_{\text{PNK}}$ would establish the over-claim but would not specify the correction. The canonical counter-model $\mathcal{M}_{\text{ERR}}$ simultaneously establishes the non-derivability *and* prescribes the minimal Additive Correction — a general consequence of Theorem 7.3.

Observation 9.8 (The Architectural DAG Is Irreducible). The Architectural DAG of the Pre-Numeric Kernel has 9 vertices and 13 edges. Removing any vertex breaks the connectivity required for Brick 015 to derive its theorems; removing any edge removes a dependency genuinely required by the dependent Brick. The DAG is irreducible — a structural consequence of BOMA-compliance: *every BOMA-compliant development satisfying Minimal-Brick Construction produces an irreducible Architectural DAG.*

Observation 9.9 (The Corrected Development Prescribes Milestone 002). The canonical counter-model prescribes exactly two new Non-Conservative Bricks for Milestone 002:

Brick_016 : $\forall x : \mathbf{E}. \neg \text{Dist}(x, x)$ (Irreflexivity, $\text{ES}() = \text{FN}$),

Brick_017 : $\forall x, y : \mathbf{E}. \text{Next}(x) = \text{Next}(y) \rightarrow x = y$ (Injectivity, $\text{ES}() = \text{FN}$).

These two injections are independently necessary (neither derivable from the other plus prior content) and jointly sufficient to rule out $\mathcal{M}_{\text{ERR}}$.

10 Relation to BOMA Paper I

10.1 What This Paper Extends

This paper extends Paper I in four directions, none contradicting its results.

Extension 1 — Operational Specification. Paper I defined the structural components; this paper defines the operational properties those components jointly produce when correctly applied.

Extension 2 — Finer Epistemic Granularity. The binary Forced Step / Genuine Choice of Paper I is replaced by the four-way Epistemic Status taxonomy.

Extension 3 — Internal Diagnostic Tools. Paper I’s tools (separation formulas, structural comparison) operate between paths. The Internal Counter-Model operates within a single path, complementing those tools.

Extension 4 — Correction Procedures. Paper I established Local Consistency as the validity criterion but did not address correction procedures for flaws within a path. This paper provides those procedures through the two-mode framework of §6.

10.2 Formal Relationships

Proposition 10.1 (Refinement). *All structural results of Paper I remain valid under the four-way Epistemic Status taxonomy (Proposition 2.12). BOMA-compliance strictly strengthens Architectural Transparency (Proposition 4.3). The Architectural DAG extends the Branch Tree as its Decision-Level Projection (Definition 5.2). The Internal Counter-Model and Separation Formulas are complementary, non-redundant tools (Proposition 7.5).*

10.3 Minimal Modifications to Paper I

Two additions to Paper I are indicated by the results of this paper:

Addition 1. At the end of Paper I, Remark 4.3.11, add:

A complete four-way classification of the sources of necessity is provided in Paper II of this series, §3, where the binary distinction of the present paper is extended to a four-way Epistemic Status taxonomy. The binary classification of Forced Step and Genuine Choice is preserved as a coarsening of that taxonomy.

Addition 2. After Paper I, Definition 4.4.12, add Remark 4.4.12a (as introduced in §5):

At the level of individual Blocks, the full architectural structure may be richer than a tree: a Block may be required in multiple branches without local modification, constituting a Convergence Point. The formal treatment of Convergent Blocks and the full Architectural DAG is developed in Paper II, §6.

10.4 What Remains for Paper III

Papers I and II together establish BOMA as a theoretical framework (Paper I) with a complete operational specification (Paper II). Paper III will provide a fully developed mathematical example:

1. **Milestone 002 Completion:** introduce Bricks 016 (irreflexivity) and 017 (injectivity) as the prescribed Additive Correction;

2. **Construction of N:** formal development within BOMA, building on the corrected Pre-Numeric Kernel;
3. **First Branch Development:** open a Branch at the Brick 011 Decision Point and identify the first non-trivial Convergent Blocks between the two paths;
4. **Architectural Metrics:** track the growth of the DAG, Branchability Index, Conservativity Ratio, and cumulative injection weight as the development matures.

11 Conclusion

11.1 What This Paper Has Not Done

This paper has not proposed a new mathematical theory, revised Paper I, or completed the Transparent Mathematics project’s Milestone 002. The Additive Correction prescribed by ERR-001 has been specified and analyzed, but not introduced as formal Bricks. The analysis of §9 is diagnostic, not constructive.

11.2 What This Paper Has Done

We have formalized BOMA-compliance as a conjunction of six properties (Definition 4.1), enriched the epistemic classification to four-way status (§2), introduced the mandatory Conservativity Declaration (§3), extended the Branch Tree to the Architectural DAG and introduced Convergent Blocks as the architectural criterion for foundational invariance (§5), distinguished two modes of correction (§6), formalized the Internal Counter-Model as a structural prescription (§7), specified the Machine-Readable Block Record (§8), and demonstrated all contributions through a complete worked example (§9).

11.3 The Two Central Claims

First claim: BOMA-compliance is not merely a theoretical ideal but a practicable discipline. The Pre-Numeric Kernel satisfies five of the six compliance properties fully and the sixth substantially, through an independently motivated organizational discipline that converged on BOMA-compatible structure without prior knowledge of the framework.

Second claim: The Convergent Block is the precise architectural criterion for foundational invariance. A mathematical result that appears identically in two paths diverging at a Decision Point is path-independent with respect to that Decision Point — invariant under the foundational choice made there. This is the formal expression

of “foundationally neutral” mathematics: not a philosophical characterization but a structural certificate, computable from the Architectural DAG, establishing precisely which results survive which foundational choices.

11.4 A Remark on Mathematical Organization

This paper has argued throughout that how mathematics is organized is not epistemically neutral. The organizational paradigm of a mathematical development determines what is visible, traceable, correctable, and comparable. The Transparent Mathematics project demonstrates that a development can be simultaneously mathematically rigorous (mechanically verified in Lean 4) and architecturally transparent (Bricks documented, Epistemic Status declared, ERRATA maintained). The compatibility requires discipline, infrastructure, and a commitment to recording the construction history alongside the mathematical content. But it is achievable — from the very beginning of a foundational development.

The architecture is open. The Convergent Blocks are waiting to be discovered. The Branches are waiting to be developed. And the mathematical territory beyond the Pre-Numeric Kernel is structurally present, determinately reachable, and waiting to be made actual by the Bricks that will traverse it.

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